

Harindu Bandara



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0009-0009-3360-8980 | BSc | Munich, Germany

M.Sc. student at TUM specializing in Mechatronics, Bionics and Robotics. Experienced in R&D, engineering design, and academic instruction.

Education

- Master of Science in Mechatronics, Robotics and Biomechanical Engineering**
TECHNICAL UNIVERSITY OF MUNICH
2025 - Present
Munich, Germany
- Member of Enhands
 - Candidate at TUM Venture Labs
- Bachelor of Science in Mechanical Engineering**
UNIVERSITY OF MORATUWA
2020 - 2024
Katubedda, Sri Lanka
- GPA: 3.52 | German equivalent: 1.7
 - Dean's List: Semester 8
 - Research: Development of a Trans-radial prosthetic device
 - Major Coursework: Robotics | Biomedical Engineering | Control Systems | Computer-Aided Engineering
 - Extra-curricular: Karate | Classical Music Society
- CIMA Certificate Level**
WISDOM BUSINESS ACADEMY
2019
Colombo, Sri Lanka
- Financial Accounting | Management Accounting | Economics | Business Law
- G.C.E. Advanced Level**
ANANDA COLLEGE
2016 - 2018
Colombo, Sri Lanka
- Member of: Science Society | IT Society
- G.C.E. Ordinary Level**
VIDURA COLLEGE
2004 - 2015
Kotte, Sri Lanka
- Extra-curricular: Scouting

Professional Experience

- Temporary Instructor**
UNIVERSITY OF MORATUWA
2025
Katubedda, Sri Lanka
- Designed and authored a new laboratory practical curriculum in Biosignal Processing
 - Developed comprehensive educational materials to enhance student learning outcomes
 - Conducted rigorous data collection using an optical Motion Capture system
- Biomedical Engineering Robotics Design of Machine Elements Advanced Manufacturing Mechanics of Machines Manufacturing Engineering
- Visiting Instructor**
UNIVERSITY OF MORATUWA
2024
Katubedda, Sri Lanka
- Faculty of Engineering
 - Faculty of Medicine
- Control Systems ECG Op Amps
- Design Engineer (Intern)**
ARIMAC LANKA
2023
Kollupitiya, Sri Lanka
- Worked in a team that designed the navigation and intelligence of the Diyazen robot
- ROS Python C++ AI

Publications

- [1] G. M. H. D. Bandara, D. D. Amarasinghe, H. A. N. P. Hettiarachchi, H. H. M. J. D. Silva, R. K. P. S. Ranaweera, and R. A. R. C. Gopura, "Dexterous Robotic Finger with Joint-Mobility Modulation for Adaptive Grasping," in *Moratuwa Engineering Research Conference (MERCon)*, 2024.

Conferences

- **MERCon**
UNIVERSITY OF MORATUWA
- 2024
Katubedda, Sri Lanka
- Presenting Author: Dexterous Robotic Finger with Joint-Mobility Modulation for Adaptive Grasping
 - Best paper nomination for Mechanical Systems track

Projects and Associations

- **Design of a Trans-radial Prosthetic Device**
BIONICS LAB, UNIVERSITY OF MORATUWA
- 2023 - 2024
Katubedda, Sri Lanka
- Development of a patent-pending mechanism
 - Extensive research on background and technical aspects
 - Detailed design, simulations and mathematical modelling
 - Prototype fabrication and real-world testing
 - Control system design and implementation using C++ and Python
- **Industrial Optimization Project**
LANKEM PLC
- 2024
Ja-Ela, Sri Lanka
- Field visits, data gathering
 - Analysis, optimization and process simulations
- **Development of a Heart-on-a-chip MEMS device**
DEPARTMENT OF MECHANICAL ENGINEERING
- 2023
Katubedda, Sri Lanka
- Design of a Microfluidic Chip and a Piezoelectric diaphragm
 - Flow analysis and structural analysis using COMSOL
- **Design of a gearbox for a Jeweler's Rolling Mill**
DEPARTMENT OF MECHANICAL ENGINEERING
- 2022
Katubedda, Sri Lanka
- 3D modelling of gearbox
 - Programmatic calculation of design parameters using Julia

Skills and Certificates

English	Cambridge English IELTS
Research	Rapid Research Knowledge Management
Analysis	Finite Element Dynamic Systems Control Systems
Design	3D Modelling Mechanical Design Control System Design
Writing	Research papers Patents Technical reports
AI and ML	Data preprocessing Training and Model Selection Model Application
Prog. Lang.	C++ Python Julia MATLAB JavaScript

Scholarships and Awards

- **University Scholarship Scheme**
CEYLON PETROLEUM STORAGE TERMINALS LTD.
- 2021 - 2024
- **Price for Academic Performance**
ANANDA COLLEGE
- 2019
- **Prize for best performance in Mathematics and English**
VIDURA COLLEGE
- 2015

References

- **Prof. Ruwan Gopura**
UNIVERSITY OF MORATUWA
- Dean, Faculty of Graduate Studies
 - gopurar@uom.lk
- **Dr. Pubudu Ranaweera**
UNIVERSITY OF MORATUWA
- Senior Lecturer, Department of Mechanical Engineering
 - pubudur@uom.lk